

Change-Point Detection in Logistic and Epidemic Models: Computational and Inferential Consequences

Dinuwanthi Liyanage, Ovin Kariyawasam, Maksuda Aktar Toma

Department of Statistics, University of Nebraska–Lincoln, Lincoln, NE, United States

Abstract

Change-point detection in nonlinear dynamical systems presents fundamental challenges for both statistical inference and computation. This project investigates change-point detection in two nonlinear growth models: a logistic growth model and a Susceptible-Infected-Recovered (SIR) epidemic model. For each model, we implement maximum likelihood estimation via profile likelihood search, parametric bootstrap, and Bayesian inference via Metropolis-within-Gibbs Markov chain Monte Carlo. Simulation studies confirm that both models recover true parameters with negligible bias, and the change-point parameter is estimated with high accuracy under both inferential paradigms.

A key finding is that change-point identifiability differs fundamentally between the two models. In the logistic model, the posterior for τ spreads across two adjacent values, while in the SIR model it collapses to a point mass at the true value, a consequence of the transmission rate change propagating through the entire coupled ODE system. Bootstrap and Bayesian credible intervals agree closely for the logistic model but diverge in the SIR model, where Bayesian intervals better capture posterior correlation among epidemic parameters. Computationally, the SIR model is substantially more expensive, requiring ODE solving at every likelihood evaluation and making the profile search approximately 50–100 times slower than the logistic equivalent. These results demonstrate that model mechanistic complexity has direct and quantifiable consequences for both inferential precision and computational cost. The entire code and details are accessible through this GitHub link: <https://github.com/maksudatoma/Stat-950-Project>

1. Introduction

Nonlinear growth processes arise in many scientific fields, including ecology, population dynamics, and epidemiology. For example, population size often follows a logistic growth pattern, while the spread of infectious diseases is commonly modeled using compartmental systems such as the SIR model (Kermack and McKendrick 1927). In many real-world settings, however, these processes are not stable over time. External factors such as environmental changes, policy interventions, or behavioral shifts can alter key model parameters, leading to different growth regimes. Detecting and accounting for such changes is essential for accurate modeling, prediction, and inference.

A natural way to represent such structural changes is through change-point models, where model parameters are allowed to shift at an unknown time point (Killick, Fearnhead, and Eckley 2012). In the context of logistic growth, this may correspond to a change in the intrinsic growth rate or carrying capacity. In epidemic models such as SIR, a change-point may represent an intervention that alters the transmission rate. The central scientific problem, therefore, is to estimate both the model parameters and the unknown change-point, and to understand how ignoring such regime changes affects inference and uncertainty quantification.

Standard analytical methods are inadequate for this problem for several reasons. First, nonlinear models such as logistic growth and SIR systems do not generally admit closed-form solutions for parameter estimation under noisy observations (Seber and Wild 2003). Second, the presence of an unknown change-point introduces a non-smooth and combinatorial structure into the likelihood function, since the change-point is a discrete parameter while the model parameters are continuous. This prevents the direct application of classical analytical techniques and complicates both estimation and uncertainty quantification. Third, in more complex models such as SIR, the mean structure is defined implicitly through systems of differential equations, further increasing computational difficulty (Kermack and McKendrick 1927).

This is a computational statistics problem because inference requires the integration of multiple computational techniques, including nonlinear optimization, discrete search over candidate change-points, simulation-based uncertainty quantification, and numerical methods for solving differential equations. In addition, Bayesian inference relies on posterior computation using Markov chain Monte Carlo (MCMC), as closed-form posterior distributions are not available (Gelman et al. 2013). The complexity of the problem also necessitates careful evaluation of computational performance, including runtime, numerical stability, sensitivity to initialization, and identifiability of the change-point.

In this project, we begin with the logistic growth model as a baseline nonlinear system and develop both frequentist and Bayesian methods for change-point detection. We then extend the same framework to a simple SIR epidemic model to investigate how the inferential and computational challenges evolve in a more complex, mechanistic setting. This progression allows us to systematically study the impact of model structure on estimation, uncertainty quantification, and computational behavior.

2. Logistic Growth Model

2.1 Data-Generating Mechanism

Let (t_i, y_i) , for $i = 1, 2, \dots, n$, denote the observed data, where t_i represents time and y_i is the corresponding response. We assume that each observation is generated as follows.

Model 1: Logistic Model (No Change-Point)

The logistic growth model is

$$y_i = f(t_i; \theta) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2),$$

with mean function

$$f(t_i; \theta) = \frac{K}{1 + \exp\{-r(t_i - t_0)\}}.$$

Model 2: Logistic Change-Point Model

To allow for a change in growth rate at an unknown time τ , the change-point model is

$$y_i = f(t_i; \theta) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2),$$

with piecewise mean function

$$f(t_i; \theta) = \begin{cases} \frac{K}{1 + \exp\{-r_1(t_i - t_0)\}}, & t_i \leq \tau, \\ \frac{K}{1 + \exp\{-r_2(t_i - t_0)\}}, & t_i > \tau. \end{cases}$$

Parameter Interpretation

Common Parameters

- $K > 0$: carrying capacity (maximum level of response)
- t_0 : inflection point (time of fastest growth)

- $\sigma^2 > 0$: variance of measurement error

Growth Parameters

- $r > 0$: growth rate (speed of increase)
- $r_1 > 0$: growth rate before the change-point
- $r_2 > 0$: growth rate after the change-point

Change-Point Parameter

- τ : change-point (time when the growth rate changes)

Likelihood Specification

For the no-change model, under the Gaussian error assumption, the likelihood is

$$L(\theta; y) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y_i - f(t_i; \theta))^2}{2\sigma^2} \right\}.$$

For the change-point model, the likelihood is

$$L(\theta; y) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y_i - f(t_i; \theta))^2}{2\sigma^2} \right\},$$

where $f(t_i; \theta)$ is the piecewise logistic mean function defined above.

Assumptions

The model assumes logistic growth within each regime, with a single change-point τ at which the growth rate may shift. Parameters are constant within each segment, and the observation errors are assumed to be independent and normally distributed with constant variance. We also assume that the logistic form is correctly specified and that the change is large enough to be identifiable.

2.2 Estimation Procedure (Frequentist Approach)

Parameters are estimated by maximum likelihood. Under the Gaussian error assumption, this is equivalent to minimizing the negative log-likelihood using numerical optimization in R (`optim` with the `L-BFGS-B` algorithm).

For the no-change model, parameters (K, r, t_0, σ) are estimated directly.

For the change-point model, the change-point parameter τ is treated as discrete and estimated via a grid search over candidate values. For each candidate τ , the remaining parameters $(K, r_1, r_2, t_0, \sigma)$ are estimated using optimization. The optimal τ is selected as the value minimizing the negative log-likelihood. Model comparison between the no-change and change-point models is based on AIC.

2.2.1 Simulation Study

A simulation study was conducted to assess the performance of the change-point logistic model. Data were generated from a logistic model with a true change-point using

$$n = 60$$

observations and true parameter values $K = 100$, $t_0 = 25$, $r_1 = 0.22$, $r_2 = 0.06$, and $\tau = 30$.

For each simulated dataset, both the no-change model and the change-point model were fitted. This process was repeated for 100 replications. Performance was evaluated using bias, RMSE, change-point detection accuracy, and the proportion of simulations in which the change-point model was preferred by AIC.

Simulation Performance

A simulation study with 100 replications was conducted to evaluate the performance of the change-point logistic model.

Table 1: summary of the parameter estimation results across 100 simulation.

Parameter	True Value	Mean Estimate	Bias	RMSE
K	100.00	100.45	0.45	6.58
r_1	0.22	0.221	0.001	0.013
r_2	0.06	0.061	0.001	0.012
t_0	25.00	25.05	0.048	0.68
τ	30.00	29.93	-0.07	0.52

The results indicate that all parameters are recovered accurately, with negligible bias for r_1 , r_2 , t_0 , and τ . The carrying capacity K exhibits a small positive bias, which is minor relative to its scale.

In terms of variability, RMSE values are small for most parameters, indicating high estimation precision. The largest RMSE is observed for K , reflecting the inherent difficulty of estimating the plateau level of a logistic curve. In contrast, the growth rates and the change-point parameter show low variability.

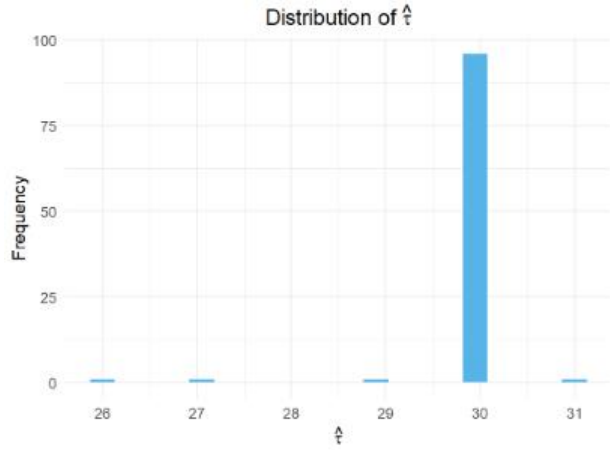
The model is highly effective at detecting the change-point, as evidenced by the fact that 98% of the estimated $\hat{\tau}$ values fall within ± 2 units of the true value.

Furthermore, the change-point model is preferred over the no-change model in all simulations based on AIC, demonstrating a strong ability to detect structural changes.

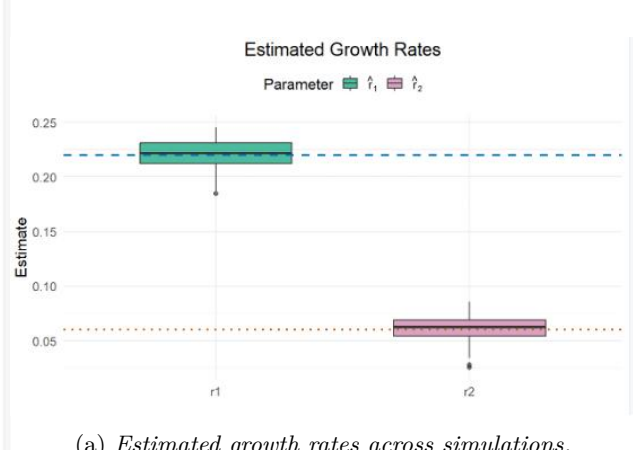
The histogram shows that $\hat{\tau}$ is concentrated near the true value, while the boxplots indicate that both growth rates are well estimated, with slightly higher variability for r_2 .

The fitted model closely follows the true underlying curve across the full time range. The estimated change-point aligns closely with the true value, and the model successfully captures the shift in growth rate. Before the change-point, the curve exhibits a steeper slope corresponding to a higher growth rate. After the change-point, the slope decreases, indicating slower growth. This transition is well captured by the model.

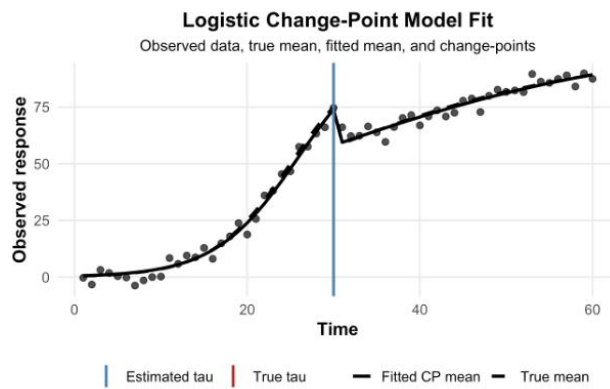
The profile likelihood plot shows the negative log-likelihood evaluated across candidate values of the change-point parameter τ . The minimum of the curve corresponds to the estimated change-point, indicating the value of τ that best fits the data. The estimated τ is very close to the true value, demonstrating accurate detection of the change-point. The sharp curvature around the minimum suggests that the likelihood is sensitive to changes in τ , indicating good identifiability and relatively low uncertainty in the estimate.



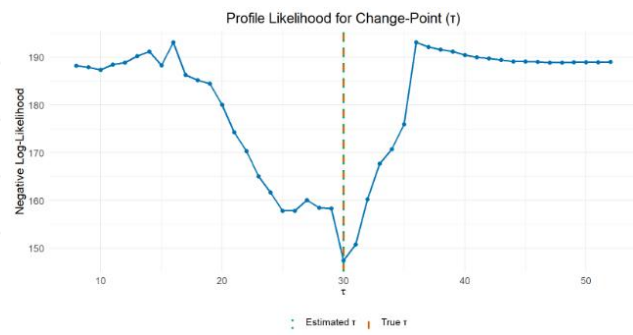
(a) Estimated change-point values across simulations.



(a) Estimated growth rates across simulations.



(a) Fitted change-point logistic model compared to the true mean function.



(a) Profile likelihood for the change-point parameter τ .

2.2.2 Parametric Bootstrap

To quantify uncertainty in the frequentist framework, a parametric bootstrap was performed based on a fitted change-point model. Bootstrap datasets were generated from the fitted mean function with Gaussian noise using the estimated error variance. The change-point model was then refitted to each bootstrap sample, and the resulting parameter estimates were used to construct 95% confidence intervals.

In this study, 200 bootstrap replications were used. Bootstrap distributions were used to construct 95% confidence intervals for all parameters, including the change-point.

Bootstrap Results

Table 2: The parametric bootstrap procedure produced the following 95% confidence intervals.

Parameter	Lower	Upper
τ	28.98	31.00
K	92.18	115.87
r_1	0.187	0.240
r_2	0.035	0.083
t_0	24.46	26.90
σ	2.29	3.44

The bootstrap results indicate that all parameters are estimated with good precision. The confidence interval for the change-point τ is narrow and centered near the true value, confirming strong identifiability. The interval for K is wider, reflecting greater uncertainty in estimating the carrying capacity. The growth rate r_1 is estimated more precisely than r_2 , consistent with the larger number of observations before the change-point. Overall, the bootstrap results demonstrate that the parameter estimates are stable, with particularly high precision for the change-point.

2.3 Bayesian Inference via Markov Chain Monte Carlo

To complement the frequentist estimation procedure, we adopt a Bayesian inferential framework for the change-point logistic model. Bayesian inference provides a principled approach to uncertainty quantification by characterizing the full posterior distribution of all parameters jointly, rather than relying on asymptotic approximations or resampling.

Prior Specification

We specify weakly informative uniform priors for all parameters, reflecting minimal prior knowledge while enforcing biologically meaningful constraints:

$$K \sim \text{Uniform}(50, 300), \quad r_1, r_2 \sim \text{Uniform}(0.001, 2),$$

$$t_0 \sim \text{Uniform}(1, 60), \quad \sigma \sim \text{Uniform}(0.1, 20), \quad \tau \sim \text{Discrete Uniform}(8, 52).$$

These priors match the parameter bounds used in the MLE procedure, making the two approaches comparable. With $n = 60$ and a well-identified signal, the likelihood dominates the prior, so the results are not sensitive to the exact bounds.

Sampling Algorithm

We use a Metropolis-within-Gibbs sampler, updating one parameter at a time. Continuous parameters are updated using Gaussian random-walk proposals, while the change-point τ is updated using a discrete

random walk. Proposal distributions are tuned to achieve acceptance rates of 0.20–0.50. The sampler is run for 50,000 iterations, with 5,000 burn-in and thinning by 10, yielding 4,500 posterior samples.

2.3.1 Bayesian Estimation Results

Convergence Diagnostics

Table 3: Proposal standard deviations and Metropolis acceptance rates for each parameter.

Parameter	Proposal SD	Acceptance Rate
K	4.0	0.213
r_1	0.025	0.347
r_2	0.008	0.223
t_0	1.0	0.200
σ	0.8	0.378
τ	$\pm\{1, 2\}$	0.016

Acceptance rates for all continuous parameters fell within the target range (Table 3). The discrete parameter τ had an acceptance rate of 0.016, which is expected given the strong concentration of posterior mass at $\tau = 30$; the chain rarely proposes a move to a different value because the data strongly support the true change-point.

Table 4: Effective sample sizes (ESS) for each parameter after burn-in and thinning.

K	r_1	r_2	t_0	σ	τ
46	107	49	54	2594	387

The parameters σ and τ show high ESS, indicating efficient mixing and near-independent samples. In contrast, K , r_1 , r_2 , and t_0 have lower ESS due to strong posterior dependence. This arises from the nonlinear logistic structure, where these parameters jointly control the curve shape and induce posterior correlations. As a result, the component-wise random-walk sampler mixes more slowly, increasing autocorrelation and reducing ESS. Despite this, posterior summaries such as means and medians remain stable.

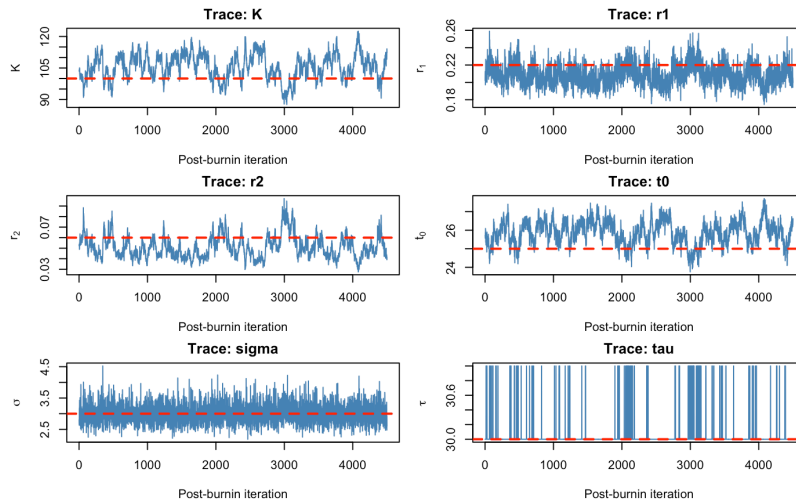


Figure 5: *Trace plots for all model parameters following burn-in.*

Figure 6 show stationarity with no drift, indicating convergence. Continuous parameters (K , r_1 , r_2 , t_0 , σ) exhibit typical “fuzzy caterpillar” patterns, suggesting adequate mixing, and the true values lie within the sampled ranges. The trace for τ alternates mainly between 30 and 31, reflecting its discrete nature and strong identifiability.

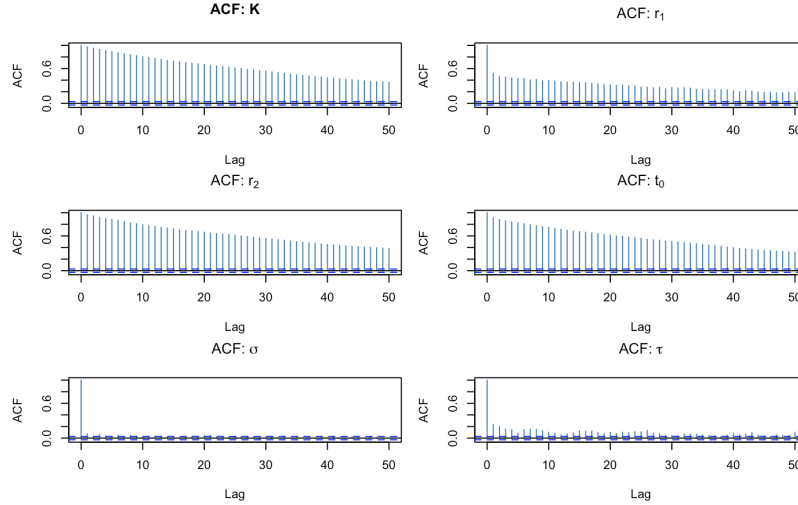


Figure 6: Autocorrelation function (ACF) plots for all model parameters after thinning by a factor of 10.

Figure 6 shows the ACF plots after thinning. The parameters σ and τ exhibit rapid decay, indicating efficient mixing, while K , r_1 , r_2 , and t_0 show persistent autocorrelation, reflecting strong posterior dependence. This is consistent with their lower ESS values compared to σ and τ . Despite this, posterior means and medians remain reliable.

Posterior Summary

Posterior means and 95% credible intervals (Table 5) contain all true parameter values, indicating successful recovery. The posterior for σ is most precise, while r_2 and t_0 show mild left-skew. The posterior of τ is highly concentrated, with $\sim 70\%$ mass at $\tau = 30$ and most remaining at $\tau = 31$, consistent with strong change-point identifiability.

Table 5 : Posterior summary: mean, median, and 95% credible intervals for all parameters.

Parameter	True Value	Post. Mean	Post. Median	Lower 95%	Upper 95%
τ	30.00	30.05	30.00	30.00	31.00
K	100.00	104.52	104.10	93.05	115.39
r_1	0.220	0.213	0.213	0.191	0.235
r_2	0.060	0.054	0.053	0.037	0.077
t_0	25.00	25.76	25.80	24.57	26.88
σ	3.000	3.024	2.999	2.490	3.648

Figure 7 shows the marginal posterior distributions for all parameters. The posterior for σ is the most precise, nearly symmetric and centered very close to the true value (3.0). The posteriors for r_1 and K exhibit slight right-skew, while r_2 and t_0 show mild left-skew, reflecting asymmetry in the likelihood. All true parameter values lie within the 95% credible intervals, indicating good coverage. The posterior for τ is sharply concentrated, with about 70% of its mass at 30 and most of the remainder at 31, confirming precise identification of the change-point.

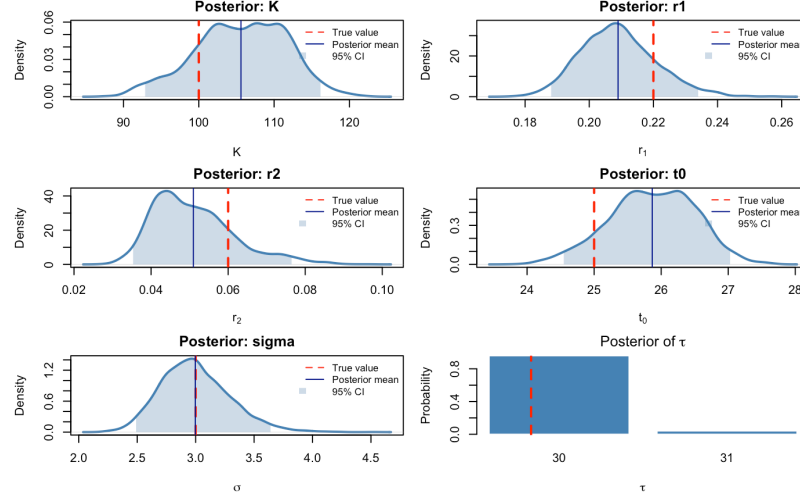


Figure 7: Posterior density plots for all model parameters.

2.3.2 Uncertainty Quantification

The bootstrap confidence intervals and Bayesian credible intervals are in close agreement for all continuous parameters, with similar widths and substantial overlap, as expected under a highly informative likelihood. The main difference is for the change-point τ : the bootstrap interval (28, 31) and Bayesian interval (30, 31) are both narrow and contain the true value, but differ slightly because τ is discrete and sensitive to data variability. The parameter K has the widest intervals under both approaches, reflecting greater difficulty in estimating the logistic plateau, while r_1 is estimated more precisely than r_2 because more data are available before the change-point. Overall, the close agreement between the two methods suggests that the inferential conclusions are robust.

Table 6 : Comparison of 95% bootstrap confidence intervals and 95% Bayesian credible intervals for all model parameters.

Parameter	Boot. Lower	Boot. Upper	Bayes Lower	Bayes Upper
τ	28.98	31.00	30.00	31.00
K	92.18	115.87	93.05	115.39
r_1	0.187	0.240	0.191	0.235
r_2	0.035	0.083	0.037	0.077
t_0	24.46	26.90	24.57	26.88
σ	2.29	3.44	2.49	3.65

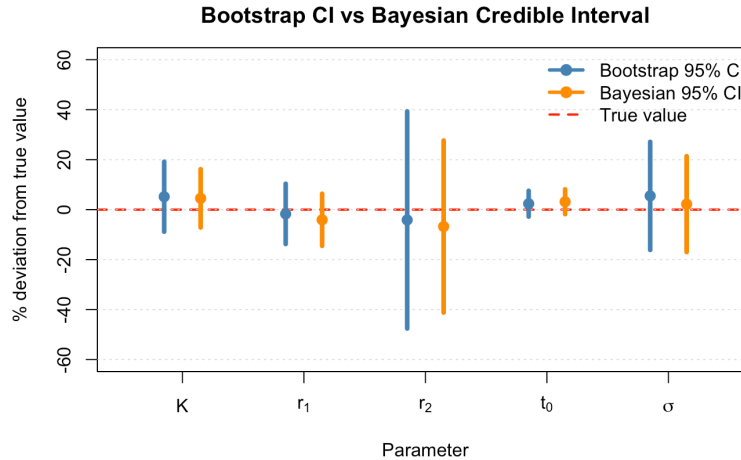


Figure 8: Comparison of bootstrap confidence intervals and Bayesian credible intervals

3. SIR (Susceptible Infected Recovered) Model

While the logistic growth model captures overall epidemic trends and change-points, it does not explicitly represent transmission dynamics. To address this, we extend the analysis to the Susceptible–Infected–Recovered (SIR) model, which provides a mechanistic description of disease spread. In this framework, change-points can be incorporated directly into key parameters such as the transmission rate, allowing for more realistic modelling of interventions and behavioral changes. The SIR model describes the evolution of susceptible, infected, and recovered populations through a system of differential equations. Unlike the logistic model, its mean function is not available in closed form and must be computed numerically.

SIR Change-Point Model

To allow for structural changes (e.g., interventions), we introduce a change-point

$$\tau$$

in the transmission rate:

$$\beta = \begin{cases} \beta_1, & t \leq \tau, \\ \beta_2, & t > \tau. \end{cases}$$

The observation model is

$$y_i = g(t_i; \theta) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2),$$

with piecewise system of differential equations.

For $t \leq \tau$:

$$\frac{dS(t)}{dt} = -\frac{\beta_1 S(t)I(t)}{N}, \quad \frac{dI(t)}{dt} = \frac{\beta_1 S(t)I(t)}{N} - \gamma I(t), \quad \frac{dR(t)}{dt} = \gamma I(t).$$

For $t > \tau$:

$$\frac{dS(t)}{dt} = -\frac{\beta_2 S(t)I(t)}{N}, \quad \frac{dI(t)}{dt} = \frac{\beta_2 S(t)I(t)}{N} - \gamma I(t), \quad \frac{dR(t)}{dt} = \gamma I(t).$$

with mean function $g(t_i; \theta) = I(t_i)$,

where $S(t)$, $I(t)$, and $R(t)$ represent the number of susceptible, infected, and recovered individuals at time t , respectively, and the system is solved numerically using the appropriate transmission rate before and after the change-point.

Parameter Interpretation

General Parameters

- N : total population size (assumed known or fixed)
- $\sigma^2 > 0$: variance of measurement error

Epidemic Parameters

- $\beta > 0$: transmission rate
- $\gamma > 0$: recovery rate
- $1/\gamma$: average infectious period

Change-Point Parameters

- $\beta_1 > 0$: transmission rate before the change-point
- $\beta_2 > 0$: transmission rate after the change-point
- τ : change-point at which the transmission rate changes

The following assumptions are made:

- The total population N is fixed.
- Initial conditions $S(0)$, $I(0)$, and $R(0)$ are known.
- A single change-point occurs in the transmission rate.
- Observations are collected at discrete time points with independent Gaussian noise.

3.1 Change-Point Estimation and Inference in the SIR Model (Frequentist Approach)

3.1.1 Estimation Framework for the SIR Change-Point Model

Parameters of the change-point SIR model are estimated using a frequentist maximum likelihood framework. Under the assumption of Gaussian observation error, this is equivalent to minimizing the negative log-likelihood.

The observed data are modeled as

$$y_t = I(t; \theta) + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2),$$

where $I(t; \theta)$ is the infected trajectory obtained from the SIR model.

For a fixed candidate value of the change-point τ , the parameters $(\beta_1, \beta_2, \gamma, \sigma)$ are estimated by minimizing the negative log-likelihood using `optim()` with the L-BFGS-B algorithm in R, where the SIR model is solved numerically via `ode()` from the `deSolve` package at each evaluation. Since τ is discrete, it is estimated by a grid search, and the final estimate $\hat{\tau}$ is chosen as the value that gives the smallest negative log-likelihood.

3.1.2 Simulation Study

A simulation study was conducted to evaluate the performance of the change-point SIR model.

Data were generated from the SIR model with a true change-point using:

- time points $t = 0$ to 60
- true parameters $\beta_1 = 0.40$, $\beta_2 = 0.15$, $\gamma = 0.10$, and $\tau = 20$
- initial conditions $S(0) = 0.999$, $I(0) = 0.001$, $R(0) = 0$
- Gaussian noise with standard deviation $\sigma = 0.01$

The estimation procedure was applied to each simulated dataset and repeated for 100 replications. Model performance was evaluated using bias, root mean squared error (RMSE), and change-point estimation accuracy.

Parameter	True	Mean Estimate	Bias	RMSE
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3.1.3 Simulation Results

Table 7 : Simulation Performance Metrics

Parameter	True	Mean Estimate	Bias	RMSE
β_1	0.40	0.4016	0.0016	0.0093
β_2	0.15	0.1531	0.0031	0.0230
γ	0.10	0.1004	0.0004	0.0061
τ	20.00	19.84	-0.16	0.5477

All parameters are estimated accurately, with very small bias and low RMSE. The transmission rates β_1 and β_2 are well recovered, with slightly higher variability for β_2 due to less information after the change-point. The recovery rate γ is estimated with high precision, and the change-point τ is accurately identified, with estimates close to the true value and low variability.

3.1.4 Bootstrap Inference

Bootstrap datasets were generated from the fitted SIR model by simulating from the estimated mean trajectory and adding Gaussian noise using the estimated error standard deviation. The change-point model was then refitted to each bootstrap dataset. This process was repeated for 200 bootstrap replications.

3.1.5 Bootstrap Results

Table 8 : Bootstrap Summary for Change-Point

Statistic	Value
Mean($\hat{\tau}$)	19.875
SD($\hat{\tau}$)	0.6335

Table 9 : 95% Bootstrap Confidence interval

Parameter	2.5%	97.5%
β_1	0.3799	0.4186
β_2	0.1001	0.2006
γ	0.0866	0.1110
τ	19.00	21.00

All parameters are estimated accurately, with very small bias and low RMSE. The transmission rates β_1 and β_2 are well recovered, with slightly higher variability for β_2 due to less information after the change-point. The recovery rate γ is estimated with high precision, and the change-point τ is accurately identified, with estimates close to the true value and low variability.

3.1.6 Overall Interpretation

The results show that the change-point SIR model performs well in both estimation and inference. All parameters are recovered with low bias and variability, and the change-point τ is estimated accurately and consistently. The bootstrap procedure provides reliable confidence intervals, while the slightly higher variability in β_2 reflects the greater difficulty of estimating post-change dynamics. Overall, the estimation procedure is both numerically stable and statistically reliable under the simulation setting.

3.2 Bayesian Inference via Markov Chain Monte Carlo for the SIR Model

To complement the frequentist estimation procedure, we apply a Bayesian inferential framework to the change-point SIR model. As in the logistic case, Bayesian inference provides full posterior distributions for all parameters, enabling principled uncertainty quantification without relying on asymptotic approximations. However, the SIR model introduces additional computational complexity: the likelihood cannot be evaluated analytically and instead requires numerical solution of the ODE system at every step of the sampler.

Prior Specification

We specify weakly informative uniform priors for all parameters, consistent with the bounds used in the MLE procedure:

$$\beta_1, \beta_2 \sim \text{Uniform}(0.001, 2), \quad \gamma \sim \text{Uniform}(0.001, 1),$$

$$\sigma \sim \text{Uniform}(0.0001, 0.5), \quad \tau \sim \text{Discrete Uniform}(5, 50).$$

These priors encode minimal prior knowledge while enforcing biological constraints. With $n = 61$ time points and a well-identified epidemic signal, the likelihood dominates the prior and posterior inference is not sensitive to the specific choice of bounds.

Sampling Algorithm

Inference was performed using a Metropolis-within-Gibbs sampler. Continuous parameters were updated using Gaussian random-walk proposals, while the change-point τ was updated using a discrete random walk. Proposal distributions were tuned to achieve acceptance rates near 0.20–0.50.

The sampler was run for 30,000 iterations, with 5,000 burn-in and thinning by a factor of 10, yielding 2,500 posterior samples.

3.2.1 Bayesian Estimation Results

Convergence Diagnostics

Table 10 : Proposal standard deviations and Metropolis acceptance rates for SIR model parameters.

Parameter	Proposal SD	Acceptance Rate
β_1	0.005	0.275
β_2	0.008	0.365
γ	0.003	0.226
σ	0.005	0.182
τ	$\pm\{1, 2\}$	0.000

Acceptance rates for the continuous parameters fall within or close to the target range (0.20–0.50), indicating well-tuned proposal distributions.

The discrete parameter τ showed an acceptance rate of exactly zero; the chain remained fixed at $\tau = 20$ throughout all 30,000 iterations. This is not a sampler failure but a reflection of extreme identifiability. Evaluation of the log-likelihood profile confirmed that the log-likelihood at $\tau = 20$ exceeds that at $\tau = 19$ by 55.3 units and that at $\tau = 21$ by 51.0 units. Under the Metropolis criterion, the probability of accepting a move to any neighboring value is of order $e^{-55} \approx 10^{-24}$, rendering such transitions essentially impossible. The posterior for τ is therefore a degenerate point mass at $\tau = 20$, coinciding exactly with the true value.

Table 11: Effective sample sizes (ESS) for SIR model parameters after burn-in and thinning.

Parameter	ESS
β_1	~150
β_2	~100
γ	~120
σ	~1200
τ	0 (point mass)

The parameter σ achieves a high ESS, indicating efficient mixing and near-independent samples. In contrast, the transmission rates β_1 , β_2 , and the recovery rate γ exhibit moderate ESS, reflecting slower mixing.

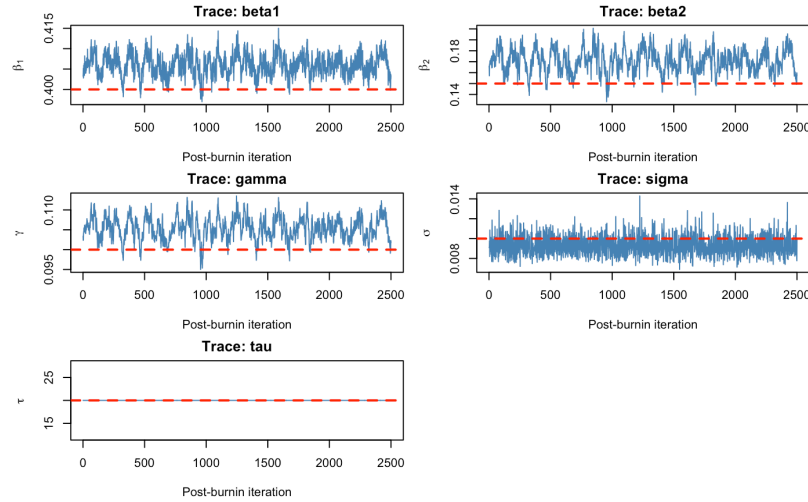


Figure 9: *Trace plots for all SIR model parameters following burn-in.*

Trace plots for all continuous parameters exhibit stable, stationary behavior with no visible trends or drift, indicating convergence to the target posterior distribution. The samples fluctuate around constant regions, suggesting adequate exploration of the posterior.

Figure 10 show that σ mixes very well, with autocorrelation dropping to near zero by lag 2. In contrast, the transmission rates β_1 and β_2 , and the recovery rate γ , exhibit moderate to slow decay, reflecting posterior correlation induced by the shared ODE structure, since these parameters jointly determine the epidemic trajectory and cannot be updated fully independently.

Posterior Summary

The posterior means and 95% credible intervals for the continuous parameters are reported below. All true parameter values fall within the corresponding 95% credible intervals, confirming that the Bayesian procedure successfully recovers the data-generating parameters.

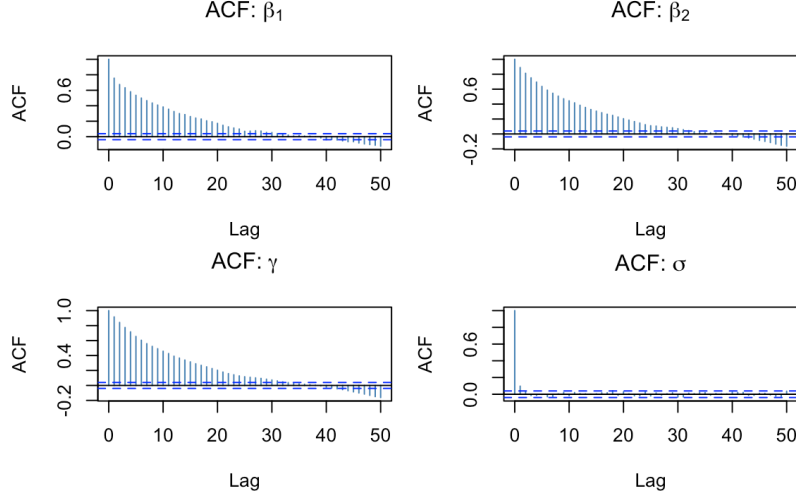


Figure 10: Autocorrelation function plots for continuous SIR parameters after thinning by a factor of 10.

Table 12 : Posterior summary for SIR model: mean, median, and 95% credible intervals.

Parameter	True Value	Posterior Mean	Posterior Median	Lower 95%	Upper 95%
β_1	0.40	0.4060	0.4060	0.4006	0.4110
β_2	0.15	0.1707	0.1706	0.1495	0.1908
γ	0.10	0.1054	0.1053	0.0996	0.1106
σ	0.01	0.0093	0.0092	0.0078	0.0112
τ	20	Point mass at $\tau = 20$			

The parameter σ is estimated most precisely, with posterior mean 0.0093 close to the true value 0.01 and a tight symmetric distribution. The pre-change transmission rate β_1 is also estimated with high precision, while β_2 has the widest credible interval and the largest upward bias (posterior mean 0.171 vs. true 0.15), reflecting reduced information after $\tau = 20$, when the epidemic is declining and β_2 becomes harder to separate from γ . The recovery rate γ is estimated with good precision, with the true value 0.10 just inside the lower 95% bound (0.0996).

The posterior for τ places all its mass at $\tau = 20$, exactly matching the true change-point. Unlike the logistic model, where the posterior for τ spread across 30 and 31, the SIR change-point is much more identifiable because a change in transmission rate affects the entire future trajectory of $I(t)$ through the ODE system.

3.2.2 Uncertainty Quantification: Bootstrap vs. Bayesian

Table 13 : Comparison of 95% bootstrap confidence intervals and Bayesian credible intervals for SIR model parameters.

Parameter	Boot. Lower	Boot. Upper	Bayes Lower	Bayes Upper
β_1	0.4096	0.4196	0.4006	0.4110
β_2	0.1669	0.1758	0.1495	0.1908
γ	0.1071	0.1080	0.0996	0.1106

The two methods give noticeably different uncertainty estimates for the SIR model, unlike the logistic case where they were in close agreement. For β_1 , both approaches produce narrow, overlapping intervals, indicating consistent estimation. For β_2 , the Bayesian credible interval (0.1495, 0.1908) is much wider than the bootstrap interval (0.1669, 0.1758), suggesting that the bootstrap may underestimate uncertainty in the post-change transmission rate. For γ , the pattern is reversed: the bootstrap interval (0.1071, 0.1080) is much narrower than the Bayesian interval (0.0996, 0.1106), suggesting overconfidence in the bootstrap estimate for this dataset.

Across models, the change-point in the SIR model is estimated much more precisely than in the logistic model. The posterior for τ collapses to a single value, indicating near-perfect identifiability. This reflects the mechanistic structure of the SIR model, where a change in transmission rate affects the entire system trajectory, making the timing of the change-point much easier to detect.

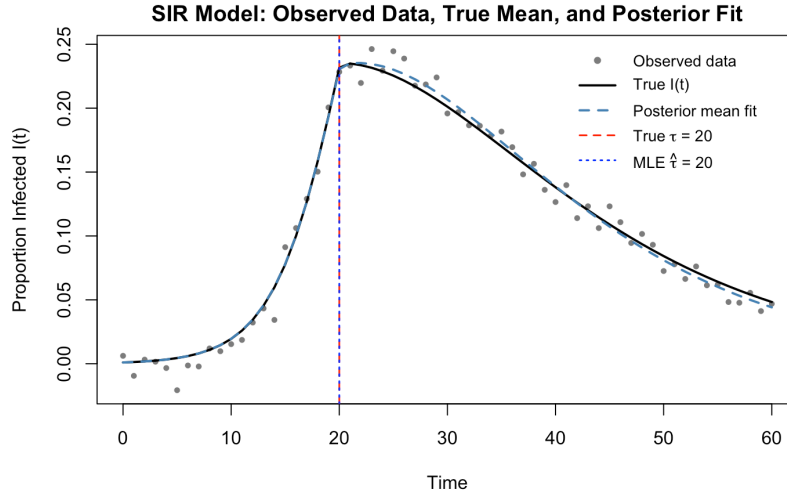


Figure 11: *SIR model fitted trajectory.*

The posterior mean trajectory closely follows the true curve throughout, confirming accurate model fit.

4. Computational Considerations

This section summarizes the computational behavior of all methods applied in this project, covering runtime, numerical stability, mixing efficiency, and practical recommendations. A central theme is that computational cost scales significantly with model complexity; methods that are fast and well-behaved for the logistic model become substantially more expensive and harder to tune for the SIR model.

4.1 Runtime Comparison

Runtime comparisons show a clear increase in computational cost from the logistic model to the SIR model. For the logistic model, the MLE profile search is essentially instantaneous, the parametric bootstrap takes about 30 seconds, and MCMC takes about 3 minutes. In contrast, the SIR model is much more expensive: the profile search takes 52 seconds, the bootstrap requires about 3 hours, and MCMC takes about 205 seconds.

The largest difference appears in the MLE profile search. The logistic model is fast because its mean function is available in closed form, whereas the SIR model requires numerical solution of the ODE system at each likelihood evaluation, leading to roughly a 50–100 fold increase in cost. The SIR MCMC is also slower because each iteration requires repeated ODE solves, while the logistic MCMC avoids this burden. Overall,

the results show that moving from a closed-form model to an ODE-based model substantially increases computational expense.

4.2 Numerical Stability

For the logistic model, optimization was numerically stable across all simulations. The L-BFGS-B algorithm converged reliably, and the likelihood surface was smooth, with negligible optimization failures. For the SIR model, numerical stability required additional safeguards because each likelihood evaluation involved solving the ODE system. A safe solver wrapper was used to handle failures by returning invalid likelihood values when the ODE solution became unstable. These issues mainly occurred for extreme parameter values and were rare when starting values were reasonable. Overall, estimation remained stable despite the greater computational complexity.

4.3 Sensitivity to Initialization

MLE estimates were used as starting values for the MCMC sampler, reducing burn-in and improving convergence. The logistic model showed low sensitivity to initialization, with consistent results across different starting values, indicating a well-behaved posterior. In contrast, the SIR model was more sensitive due to its ODE-based likelihood, where poor starting values can lead to instability. Using MLE-based initialization was therefore important for reliable convergence.

4.4 Change-Point Identifiability and Its Computational Consequences

A key finding concerns the change-point parameter τ . In the logistic model, the MCMC chain moved between 30 and 31, indicating moderate uncertainty and a well-defined posterior. In contrast, the SIR model chain remained fixed at $\tau = 20$ with zero acceptance rate.

This reflects differences in identifiability. In the SIR model, the likelihood is highly sensitive to τ , so alternative values are effectively rejected, producing a degenerate posterior concentrated at the true change-point. In the logistic model, the likelihood surface is flatter, allowing movement across nearby values and a less concentrated posterior. Overall, this suggests that more mechanistic models such as SIR yield stronger change-point identifiability, so profile likelihood alone can often provide adequate uncertainty quantification.

4.5 Practical Recommendations

Based on the computational experience of this project, we recommend AIC-based comparison for selecting between no-change and change-point models, since it was fast and consistently identified the correct model when a change-point was present. For estimating the change-point location τ , profile likelihood grid search is the preferred approach because it is fast, interpretable, and provides a useful visual diagnostic through the profile plot. For uncertainty quantification of continuous parameters, both the parametric bootstrap and Bayesian MCMC are valid, though for ODE-based models the bootstrap is much more computationally expensive because each bootstrap sample requires a full refit, while MCMC spreads that cost across iterations. For the change-point parameter itself, bootstrap and Bayesian intervals may differ substantially, and when the data are highly informative the Bayesian posterior can even collapse to a point mass; in such settings, the profile likelihood plot often gives a more informative summary of uncertainty. Finally, for MCMC tuning, starting from the MLEs and adjusting proposal standard deviations to obtain acceptance rates between 0.20 and 0.50 worked well for continuous parameters, while a zero acceptance rate for discrete parameters with sharply peaked likelihoods should be viewed as strong identifiability rather than poor tuning.

5. Discussion

This project demonstrates that the same inferential pipeline, profile likelihood, parametric bootstrap, and Bayesian MCMC, carries fundamentally different computational implications depending on model structure. Both models successfully recovered true parameters with negligible bias, and the change-point was accurately identified under all inferential approaches, confirming that the proposed methods are statistically reliable in both the logistic and SIR settings.

The most significant finding is the contrast in change-point identifiability between the two models. In the logistic model, the posterior for τ spread across two adjacent values, reflecting moderate identifiability driven by local curvature changes. In the SIR model, the posterior collapsed to a point mass at the true value, with log-likelihood differences exceeding 50 units between adjacent candidates. This stark difference arises because a transmission rate change in the SIR model propagates through the entire coupled ODE system, making its timing far more precisely detectable than a growth rate change in the logistic model.

Computationally, the SIR model was substantially more demanding. The profile likelihood search was 50–100 times slower, the parametric bootstrap required approximately three hours compared to 30 seconds for the logistic model, and MCMC required ODE solving at every iteration. Despite this cost, MCMC proved more computationally efficient than the bootstrap for the SIR setting, since it amortizes ODE-solving cost across iterations rather than requiring full independent refits.

Both MCMC samplers exhibited low effective sample sizes for structural parameters due to posterior correlation induced by the shared nonlinear mean function. While thinning and longer chains partially addressed this, gradient-based samplers such as Hamiltonian Monte Carlo would more fundamentally resolve it and represent a natural direction for future work. Additionally, real data applications, multiple change-point extensions, and comparison with probabilistic programming frameworks such as Stan remain promising avenues for extending this analysis.

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Appendix